

Monitoring and forecasting for disease and pest in crop based on WebGIS system

Yingying Dong

*Key laboratory of Digital Earth
Science, Institute of Remote Sensing
and Digital Earth, Chinese Academy of
Sciences
Aerospace Information Research
Institute, Chinese Academy of
Sciences
Beijing, China
dongyy@radi.ac.cn*

Fang Xu

*Beijing Advanced Innovation Center for
Imaging Technology, Capital Normal
University
Beijing, China
aaronxf1314@163.com*

Linyi Liu

*Key laboratory of Digital Earth
Science, Institute of Remote Sensing
and Digital Earth, Chinese Academy of
Sciences
Aerospace Information Research
Institute, Chinese Academy of
Sciences
Beijing, China
liuly35@radi.ac.cn*

Xiaoping Du

*Key laboratory of Digital Earth
Science, Institute of Remote Sensing
and Digital Earth, Chinese Academy of
Sciences
Aerospace Information Research
Institute, Chinese Academy of
Sciences
Beijing, China
duxp@radi.ac.cn*

Huichun Ye

*Key laboratory of Digital Earth
Science, Institute of Remote Sensing
and Digital Earth, Chinese Academy of
Sciences
Aerospace Information Research
Institute, Chinese Academy of
Sciences
Beijing, China
yehc@radi.ac.cn*

Wenjiang Huang*

*Key laboratory of Digital Earth
Science, Institute of Remote Sensing
and Digital Earth, Chinese Academy of
Sciences
Aerospace Information Research
Institute, Chinese Academy of
Sciences
Beijing, China
huangwj@radi.ac.cn*

Yining Zhu*

*Beijing Advanced Innovation Center for
Imaging Technology, Capital Normal
University
Beijing, China
ynzhu@cnu.edu.cn*

Abstract—Infected areas and damage levels due to pest and disease have been growing seriously according to the global climate changes. The government department of plant protection normally collects the crop pest and disease information by manual inspection, which is unable to provide timely and spatially continuous crop growth status and pest / disease development over large areas. The manuscript aims to bring together and produce cutting edge research to provide crop pest and disease monitoring and forecasting information, integrating multi-source (Earth Observation-EO, meteorological, entomological and plant pathological, etc.) to support decision making in sustainable management of pest and disease. Taking national disease -- a fungal disease of wheat rust and national pest -- a serious insect pest locust as the experimental object, we conducted the following research: 1) the sensitive spectral features of rust and locust with capability of stresses differentiation would be identified or formed based on field hyperspectral data and UAV hyperspectral images for detection, 2) multi-sources of data that are composed by remote sensing images (eg. GF, Landsat and Sentinel) and meteorological observations would be integrated to evaluating habitat of rust and locust in farmland level, 3) multi-temporal remote sensing images and vegetation ecological dataset are combined to provide environmental information of rust and locust dispersion, and forecast the damaged areas and levels in regional scales. Moreover, an automatic system is developed to do the disease and pest timeseries monitoring and forecasting, also the visual display of the thematic maps and analysis reports.

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The system is constructed based on WebGIS platform. It selected Browser/Server (B/S) architecture, access the system interface through a web browser, simple, quick and easy to operate. And, it could timely, efficiently, quantitatively do the extraction and analysis of crop pest and disease occurrence and developing information, also produce pest and disease thematic maps and scientific report. The crop pest and disease monitoring and forecasting system, can provide effective information of pest and disease developing for our agricultural sector, provide a scientific basis to formulate pest and disease prevention and control measures, and also provide data basis and technical support for the crop network management. Based on the system, we analysed the damaged situation and changes of national wheat rust in 2019, and locust in Tianjing 2019. The results would not only promote efficacy of pest and disease management and prevention by improving the accuracy of monitoring and forecasting, but also help to reduce the amount of chemical pesticides, which could thus guarantee the food security and sustainable development of agriculture in China.

Keywords—*crop, pest, disease, monitoring, forecasting, system*

I. INTRODUCTION

Effectively and timely monitoring and forecasting of regional crop pest and disease is very important for field management to enhance food security. Traditional manual field survey method for crop pest and disease monitoring could achieve detailed information of the pest or disease and its damaged characteristics, but couldn't get the regional situation, such as damaged areas and levels, also the next

dispersal trend. Remote sensing technologies as a method to timely achieve regional observations is a better way to provide data for pest and disease monitoring and development forecasting. Also, other source of data, such as meteorological, entomological and plant pathological are combined with remote sensing observation to do the monitoring and forecasting. Existed researches has done lots of work on the spectral indices construction and extraction for pest and disease monitoring, such as sensitive bands, vegetation indices, and time series changing characteristics [1-3]. Also, monitoring and forecasting models construction and validation. But, most of them pay attention on the leaf and canopy levels, couldn't applied to regional level. In order to improve the efficiency of crop pest and disease monitoring and forecasting on regional level, some researchers have made some achievements in the crop disease and pest monitoring and prediction system construction. Shelestov et al. built the agricultural monitoring information system with remote sensing satellite data and products, has the function of automatically retrieving crop information [4]. Oki et al. constructed agricultural monitoring system based on remote sensing image and field server camera data, can be applied to cabbage fertilizer estimation and crop yield prediction [5]. Zhang et al. selected PHP platform and chose the raw data in the field, construction of crop disease and pest warning system for short-term crop disease and pest warning [6]. Peng et al. based on GIS to build the apple disease and pest management information system, providing query function, such as the apple disease and pest information and thematic map [7]. Although they constructed some system for pest and disease monitoring and forecasting, but none of them make it to the regional and moreover national scale, and provide the development forecasting information, which is more useful in field management scheme design in agricultural production. In all, we aim to construct a system, integrated with multi-source data, pest and disease monitoring and forecasting models, thematic maps and scientific reports to support disease and pest control.

II. DATA AND METHODS

A. Study Area and Dataset

In this manuscript, national disease -- a fungal disease of wheat rust and national pest -- a serious insect pest locust are taking as the object. The study area of wheat rust is the mainland China, and for the locust, we choose Tianjing as the study area. The data used for pest and disease monitoring and forecasting include spatial data and attribute data. For spatial data, it include land-use data, geo-politic database, wheat planting area and locust area, and remote sensing data, which are MODIS surface reflectivity, Normalized difference vegetation index (NDVI), Land surface temperature (LST), and GF, Landsat, Sentinel images, are of important for disease and pest occurrence monitoring and future trends forecasting. For attribute data, it include tables, documents, and models. Tabular data mainly refer to rainfall, mean daily temperature, wind direction and wind speed, such spatial meteorological data (interpolated) are achieved from the weather station of China Meteorological Administration. Documents include the files or information of pest and disease occurrence from Ministry of Agriculture and Rural Affairs of China. Models involve the wheat rust and locust remote sensing monitoring and forecasting models, which developed by our team [8-10].

B. Crop Pest and Disease Monitoring and Forecasting Models

For wheat rust, it forms yellow stripe or oval-shaped spots on the leaves of wheat, which results in leaf yellowing, chlorophyll content and water content reduction. In this manuscript, the standard for wheat stripe rust is based on China's 'Rules for investigation and forecasting of wheat stripe rust' (GB/T15795-2011) [11], and the detail standards is listed in Table 1. Based on the standards, the ground survey data and remote sensing observation data were obtained through the integrated experiment of crop diseases on satellite ground synchronization. Additionally, the levels 1 and 2 in Table I refer to the slight occurrence of wheat rust, the level 3 refers to moderate, and levels 4 and 5 refer to severe.

TABLE I. WHEAT RUST FIELD INVESTIGATION INDICES

Index	Level				
	1	2	3	4	5
Disease index	0.001<Y≤5	5<Y≤10	10<Y≤20	20<Y≤30	Y>30
The rate of disease field / %	1<R≤5	5<R≤10	10<R≤20	20<R≤30	R>35

Y is the disease index, which is used to reflect the severity of disease occurrence. The calculation formula is $Y=F \cdot D \cdot 100$, F is the disease leaf rate, and D is the average severity of the disease leaf. R is the disease field rate, referring to the percentage of the number of damaged fields in the total field [11].

Firstly, the spatial distribution of wheat rust is taken as prior knowledge based on literature research and ground survey. Then Wheat Rust Index (WRI) is constructed based on Plant Senescence Reflectance Index (PSRI) (1) and Red-edge Vegetation Stress Index (RVSI) (2) to monitor wheat stripe rust, for which WRI (3) considering crop growth, chlorophyll content and their variation characteristics. At last, integrated with disease habitat information including land surface temperature (LST, MODIS product), rainfall and wind (meteorological data), also historical data to construct Disease Index (DI) (4) for wheat rust monitoring [12].

$$PSRI = (R_R - R_B) / R_{NIR} \quad (1)$$

$$RVSI = ((R_{712} + R_{752}) / 2) - R_{732} \quad (2)$$

$$WRI = f(\Delta PSRI, \Delta RVSI) \quad (3)$$

$$DI = g(WRI, LST - LST_{avg}, R - R_{avg}, W) \quad (4)$$

Where, R_R , R_B and R_{NIR} are the reflectance of red, blue, and near infrared (NIR) band, R_{712} , R_{752} and R_{732} are the reflectance of 712nm, 752nm and 732nm band. LST is land surface temperature while LST_{avg} is the average historical land surface temperature. R is rainfall while R_{avg} is the average historical rainfall. W is wind direction and speed. f and g are constructed with regression analysis based on filed survey datasets. The data range of DI is 0~100%. When $0 < DI \leq 30\%$, the damaged level of stripe rust is slight, $30\% < DI \leq 60\%$ is moderate, $DI > 60\%$ is severe.

In this manuscript, the standard for locust is based on China's 'Technical Specification for the Investigation and Forecast of *Locusta migratoria manilensis* (Meyen)' (GB/T 15799-2007) [11], and the detail standards is listed in Table 2. Based on the standards, the ground survey data and remote sensing observation data were obtained through the integrated experiment of pest on satellite ground synchronization. Additionally, the levels 1 and 2 in Table II refer to the slight

occurrence of locust, the level 3 refers to moderate, and levels 4 and 5 refer to severe.

TABLE II. LOCUST FIELD INVESTIGATION INDICES

Index	Level				
	1	2	3	4	5
Index of occurring (L)	$L \leq 0.1$	$0.1 < L \leq 0.25$	$0.25 < L \leq 0.56$	$0.56 < L \leq 0.9$	$L > 0.9$
The rate of damaged area (% , P)	$P \leq 30$	$30 < P \leq 50$	$50 < P \leq 70$	$70 < P \leq 90$	$P > 90$
Average density of the oriental migratory locust larva (head/m ² , d)	$0.2 \leq d < 0.3$	$0.3 < d \leq 0.5$	$0.5 < d \leq 0.8$	$0.8 < d \leq 1$	$d > 1$

Damaged levels of locust decided by the index of occurring (L) [11].

Firstly, the spatial distribution of locust is taken as prior knowledge based on literature research and ground survey. Then Locust Index (LI) is constructed based on NDVI (5) and Modified Soil Adjusted Vegetation Index (MSAVI) (6) to monitor locust, for which LI (7) considering crop canopy cover, soil background information and their variation characteristics. At last, integrated with pest habitat information including land surface temperature (LST, MODIS product) and wind (meteorological data), also historical data to construct Pest Index (PI) (8) for locust monitoring. The data range of PI is 0~100%. When $0 < PI \leq 30\%$, the damaged level of armyworm is slight, $30\% < PI \leq 60\%$ is moderate, $PI > 60\%$ is severe.

$$NDVI = (R_{NIR} - R_R) / (R_{NIR} + R_R) \quad (5)$$

$$MSAVI = ((2R_{NIR} + 1) - (\sqrt{((2R_{NIR} + 1)^2 - 8(R_{NIR} - R_R))})) / 2 \quad (6)$$

$$LI = f(\Delta NDVI, \Delta MSAVI) \quad (7)$$

$$PI = g(LI, LST - LST_{avg}, W) \quad (8)$$

III. SYSTEM DESIGN AND IMPLEMENTATION

Based on the data and models, we aim to construct a system for regional and national crop pest and disease monitoring and forecasting to timely produce thematical maps and scientific report to show the damaged areas and levels according to wheat rust and locust.

A. System Framework

Crop pest and disease monitoring and forecasting system selects specification development platform compliant with Java 2 Platform, Enterprise Edition (J2EE) to build a three-layer Browser/Server (B/S) network architecture, that is, the client layer, application layer and data layer. The system framework is shown in Fig. 1.

1) *Client layer*: as the Web browser provides the system human-computer interaction interface, which responsible for responding to the user's instruction, and is used to complete the user-level interaction. Including handling the response and display of the interface control interface, displaying the result of crop pest and disease remote sensing quantitative parameter inversion, providing users with monitoring and forecasting information, and exporting system products and services.

2) *Application layer*: as the Web server is the hub of system operation, which responsible for receiving and processing HTTP request sent by the browser (client layer). By receiving and analyzing user requests, and selecting the corresponding server processing, ArcGIS Server is used to process requests related to the display of geographic information, and model operation requests is set up by the Tomcat server Spring MVC framework. According to the contents of the request targeted to call study area surface reflectivity image data, the quantitative inversion of surface parameters and pest and disease monitoring and forecasting model. Finally, return the results to the user interface for display.

3) *Data layer*: mainly used to store data, including field survey, temporal and spatial continuous multi-source and multi-scale remote sensing images, meteorological data, various functional models, and products and services of the system.

In this B/S architecture, the three parts are interrelated. The client layer collects user requests and passes them to the application layer. And then the application layer according to user needs retrieve the corresponding data and models from the data layer to calculate the results, the results are stored in the data layer. At last, the application layer feeds back the response of the user requests to the client layer.

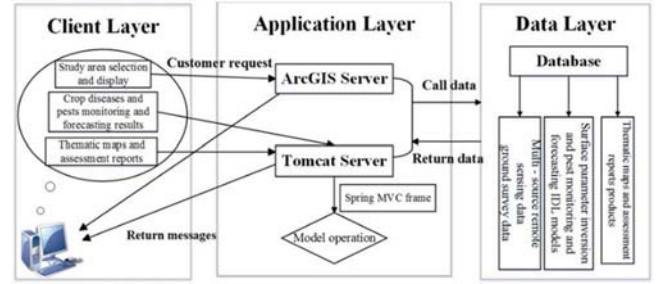


Fig. 1. Framework of crop pest and disease monitoring and forecasting system

B. System Function

The functions of crop pest and disease monitoring and forecasting system include basic function and application function. The basic function includes view zooming, roaming, and moving, information query, operation help, and other basic map operation functions. The application function includes study area selection, parameter inversion, pest and disease monitoring and forecasting, and products. This system integrates Internet, Remote sensing and GIS, and carries on the reasonable integration. Internet technology can achieve resource information dissemination, real-time data sharing. Remote sensing technology could track and monitor the occurrence location of pest and disease with time and space continuous dynamic, and visual display in the form of images. GIS technology has spatial data analysis and processing functions, can be timely and effective analysis and process of pest and disease related to spatial data. Crop pest and disease monitoring and forecasting system combine these technologies, achieve a timely and efficient regional quantitative pest and disease monitoring and forecasting. It convenient provides crop disease and pest information, thematic maps and scientific reports regularly, also a platform for timely monitoring and forecasting information Sharing.

1) *Basic function*: The map server in the system is displayed by ArcGIS for Server technology. It provides a wide range of Web-based GIS services to support the achievement of geographic data management, mapping, GIS editing functions in a distributed environment, and Web applications that simple to operate, easy to configure. If you want to open the map, you simply declare the map's URL in the map control. We need to open the cache map through ArcMap, with coordinate transformation, rendering and a series of basic operation processing. Publish the map to the ArcGIS for Server via the Internet, while setting the service name, general parameters, and the maximum and minimum proportion of the cache. After successful upload service, it could be in form of layer overlay display in the interface through the address. This method can be used to display the image data, such as inversion result image, monitoring and forecasting result images, which generated in the system.

2) *Application function*: In order to reduce the amount of data and improve the efficiency of the model, the system uses Java to build Spring MVC framework to realize the model online calculation. This process relies heavily on multi-source remote sensing data, monitoring and prediction models at the data layer. Spring MVC framework consists of Dispatcher servlet, Handler mapping, Handler adapter, Handler, and View resolver. There are two XML files as a necessary condition, one used to configure the Dispatcher servlet and the access model, another configuration Handler mapping, Handler adapter, Handler, and View resolver. It should be noted that the mapping and adapter to be one-to-one correspondence. At the same time jar package should be required for the Spring Framework, Java calls IDL into the project. The Dispatcher servlet controls the Handler adapter to execute the Handler, model as a Handler. Run the corresponding model and data processing, and then return the results to the Dispatcher servlet, through the View resolver parse into a real view. Users can access system through the view address, to achieve efficient call and computing model. This module export mainly products include the pest and disease monitoring and forecasting thematic map and scientific report, both of them need to operate after the pest and disease monitoring or forecasting. The system support BMP, JPG, TIFF, GIF, etc., as thematic map export format. Scientific report include the detailed occurrence condition of pest and disease and display their thematic maps. The reports can be preserved, printed, zoomed and output the report as a PDF.

C. Results

Based on the system, we calculated the damaged situation and changes of national wheat rust in 2019, and locust in Tianjing in 2019.

1) *Wheat rust case*: In 2019, due to the lower temperature and higher precipitation than previous years, rust was moderately occurred in winter wheat regions of China. Field temperature are higher than previous years. From April to May, maximum of the averaged field temperature in winter wheat regions of China reached 25.0°C, which is 1.2°C higher than previous years. Field precipitation from April to May are higher than previous years. According to the rainfall process

in western regions of Northeast China, North China, and central regions of South China, field humidity reached a suitable level for pests and diseases development. In 2019, the occurrence of yellow rust reached 0.7 million hectares. The disease first appeared in Jiangnan, Jianghuai, the southern regions of Huanghuai, Southwest China and Northwest China in early April. The full incidence period was during mid-late April to mid-May, and then the disease has spreaded to Northwest China, Jianghuai, and Huanghuai. The specific distributions and severities are shown in Fig.2 and Table III.

2) *Locust case*: Dagang district in Tianjing is a typical locust arrear in China. This area includes Dagang reservoir, Duliujian river, and Lier bay, which are wetland easy for locust to live. Also, it was frequently suffered from droughts in spring and water-logging in summer, this year it has a warm temperature than previous years, moreover with high vegetation covers, good for locust to survive and dispersal. According to the forecast, it will has 4.96 thousand hectares slight, 0.63 thousand hectares moderate, 3.19 thousand hectares severe for locust with this area. Details are shown in Fig.3.

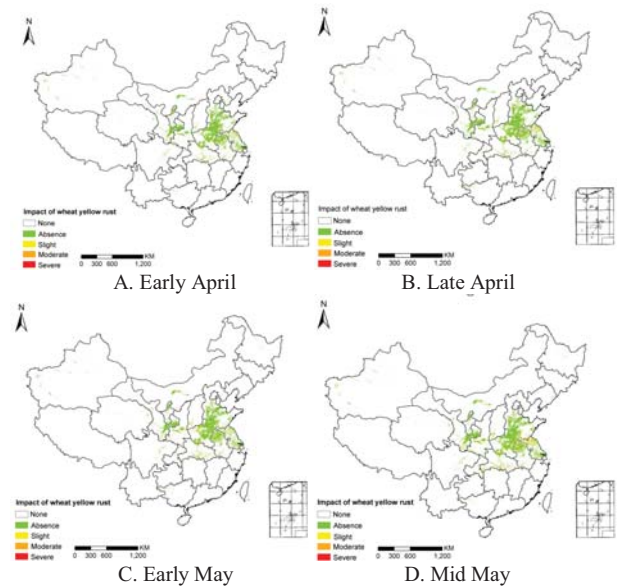


Fig. 2. Remote monitoring of wheat rust in China (2019)

TABLE III. STATISTICS OF WHEAT RUST IN CHINA (2019)

Region	Area / Thousand hectare				Total area
	Early April	Late April	Early May	Mid May	
Northeast China	0	0	0	0	85.3
North China	18	37.3	76	82	3579.3
East China	56.7	111.3	198.7	242.7	8556
South China	0	0	0	0	16.7
Central China	44	87.3	157.3	188	6710
Northwest China	22.7	45.3	82.7	96.7	3376
Southwest China	12.7	23.3	34.7	50	1828.7
Total	154.1	304.5	549.4	659.4	24152

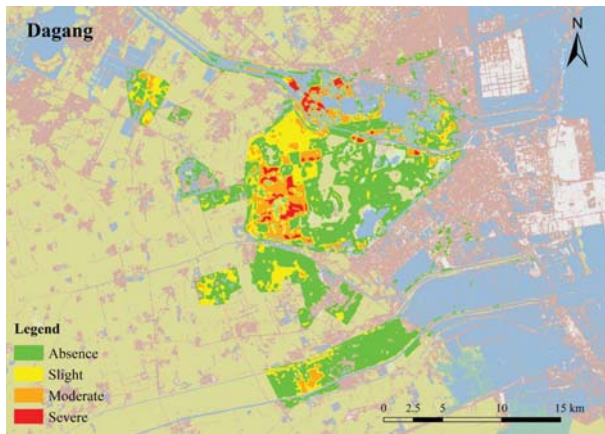


Fig. 3. Remote forecasting of locust in Tianjing (2019)

IV. DISCUSSION AND CONCLUSION

Crop pest and disease monitoring and forecasting system achieve business operation on field control and protect food security. System development includes data, calculation and production modules, each module maintains a high cohesive and low coupling relationship. Through the browser to access the system, simple, quick, and easy to operate. Using multiple types of data modeling, the accuracy is greatly improved, using Java technology integration of geographic information, remote sensing data, and models, achieving the flexibility to call model that stored in the network server for online computing. Greatly improving computing speed and efficiency, while reducing the number of data storage, realizing automatic and operational process, convenient and comfortable use of the system. The system quantitative monitor and forecast the occurrence and development of crop pest and disease. Not only for the government's agricultural sector, large-scale farms and other users to provide information on the occurrence and development of crop diseases and insect pests, but also help to improve crop yields and increase economic efficiency. It provides effective scientific basis for guaranteeing food security, and provides data and technical support for intelligent management of agricultural information. Although this system already has a complete functional system in the disease and pest monitoring and forecasting model and product export services, but in the construction of the database and improving model efficiency still need further improvement. Facing the era of large data space, endless stream of information sources. The use of data mining expertise, convenient and quick access to useful data information. Reducing the man-made participation realize automation operation flow, efficient completion of inversion,

monitoring and forecasting. Using rich reserve knowledge, to meet the different levels, different scenarios application requirements.

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